

Title : A Sustainable system to integrate Mechanical and Animal Power for Agriculture sector.

Abstract :

Agriculture not only serves as a source of livelihood in India, but it has also become a way of life. Agriculture has an impact on our fairs, festivals, and rituals. Agriculture is a time-consuming and labor-intensive operation that cannot be completed by hand. Various livestock such as bullocks and horses are utilized in agricultural fields to accomplish various activities such as ploughing, tilling, levelling, and so on. The typical arrangement is pulled in the fields by these livestock and requires a lot of energy. To make it more efficient, we propose this method, which focuses on employing only one livestock and plans to turn the livestock's energy into circular motion when the livestock works in the field. For small to medium-sized fields, an electromechanical machine can be a cost-effective and efficient way to farm. This environmentally friendly method of energy generation is used to divide the muscle strength of livestock and increase its efficiency. The output will be sent to an ISG, which will transform rotating motion into electrical energy. The electricity will be stored in a battery. As a result, it will be a renewable battery source, which can be a viable alternative.

Introduction :

As technology advances, modern equipment is added to make farming more efficient and rapid. The traditional method has aided in the conservation of natural resources, the preservation of biodiversity, and the improvement of food security. Agriculture is a time-consuming and labor-intensive operation that cannot be completed by hand. As a result, tools and machines are required to carry out agricultural procedures.

Traditional agricultural tools and implements are those that were invented in ancient times and have been used for a long time, until recently, or are still in use today. Traditional farm equipment and implements for self-sufficiency have evolved and been changed through decades of experience to address changing socioeconomic and farming issues. Traditional agricultural tools and implements were produced from locally accessible materials such as stone, wood, and iron, and were either handcrafted locally or standardised factory-made. These tools and implements were cost effective in terms of labour, money, and time.

Various livestock like bullocks, horse etc are used in the agricultural fields to perform various tasks including ploughing, tilling, levelling etc. The traditional setup is pulled in the fields by these livestock and requires a lot of energy, to make it more efficient we propose this mechanism which focuses on using just one livestock where we plan to convert livestock's energy into rotary motion when the livestock works in the field, the rotary motion obtained through this will be transferred to the gear box using a shaft where the speed modulation will take place. The output will be transferred through the generator (ISG), this will convert rotary motion into electrical energy. The electrical energy will be stored in battery. This way it will be a renewable battery source which can be an effective solution, and will help the consumer eliminate the problem of repetitive battery charging. This sustainable way of energy

generation is applied to divide the muscle power of the livestock and increase its efficiency. The mechanism also has tool replaceable feature, so that the farmer can use the system for various purposes. Hence, we avoid the use of tractors, thus, saving the environment and a sustainable way to help our farmers

Literature Review :

A non-patent literature document titled, “Smart Agricultural Machine With A Computer Vision-Based Weeding And Variable-Rate Irrigation Scheme” published in 2018 describes a scheme that combines computer vision and multi-tasking processes to develop a small-scale smart agricultural machine that can automatically weed and perform variable rate irrigation within a cultivated field. Image processing methods such as hsv (hue (h), saturation (s), value (v)) color conversion, estimation of thresholds during the image binary segmentation process, and morphology operator procedures are used to confirm the position of the plant and weeds, and those results are used to perform weeding and watering operations. Furthermore, the data on the wet distribution area of surface soil (wdas) and the moisture content of the deep soil is provided to a fuzzy logic controller, which drives pumps to perform variable rate irrigation and to achieve water savings.

A non-patent literature document titled, “Application Of Mechatronics In Agriculture: A Review” published in 2019 describes that Mechatronics has found quite a number of useful applications in agriculture. Agriculture as one of the oldest industries, dating as far back as the nomadic age originally depended solely on human effort, then apprehended animal labor, and then came mechanical advances such as diesel/steam-engine tractors and mechanical tools with hydrostatic power which needed control.[5] The answer to unresolved problems relies on more advances that necessitate the replacement of human intellect to meet the requirements for superior autonomy in more indefinite and unstructured environments. Promising disciplines in this framework include Mechatronics, Large-scale optimization and Complex system automation, and our focus is on the advancement of irrigation system. Some of the applications of mechatronics in agriculture and their processes are discussed to gain insight on the state of the art, advantages and weaknesses of several methods employed.

A non-patent literature document titled, “Hydraulic Hybrids – The New Generation of Energy Efficient Drives” published in 2009 describes use of the electronic control system which is hybrid with the mechanical and hydraulic concept. Previously, the technology was inclined only on the mechanical and hydraulic concept but the recent developments is persuaded more towards the electro mechanical concept also, to increase the efficiency, reduce the fuel consumption and automated the task and the concept has been defined as the agriculture hybrid structures. The concept also focuses, on the merged concept of mechanical energy

with the photovoltaic (PV) solar energy so that it will reduce the greenhouse gas (GHG) emissions and increase the efficiency and performance of the machine.

A non-patent literature document titled, “Electrification of Agricultural Machinery A Review” published in 2021 describes about the modern agriculture technology by integrate the electric power drives with the mechanical and hydraulic drives to vary the parameters like high power density, speed and torque control, the author developed the modern agriculture system using control system, advanced sensing system with efficient feedback controllers to increase the performance of the mechanism.

A non-patent literature document titled, “A New Concept for Seed Precision Planting” published in 2015. In this paper, the modern agriculture concept has been used to increase the efficiency of the precision planting and to increase the precision and accuracy of the sowing machine. It increases the consistency of the seed placement.

References:

- [1] Chang C. L. and Lin K. M. 2018 Smart agricultural machine with a computer vision-based weeding and variable-rate irrigation scheme *Robotics* 7 38
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- [3] Tyagi A., Gupta N., Navani J. P., Tiwari M. R. and Gupta M. A. 2017 Smart irrigation system *International Journal for Innovative Research in Science & Technology* 3
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- [5] Duckett T., Pearson S., Blackmore S. and Grieve B. 2018 Agricultural robotics: the future of robotic agriculture *Arxiv preprint arxiv:1806.06762*
- [6] Karner, J., Prankl, H., & Kogler, F. (2012). Electric drives in agricultural machinery. In *Energy, biomass and biological residues. International Conference of Agricultural Engineering-CIGRAgEng 2012: Agriculture and Engineering for a Healthier Life, Valencia, Spain, 8-12 July 2012. CIGR-EurAgEng.*
- [7] Buning, E. A. (2010). Electric drives in agricultural machinery-approach from the tractor side. *Journal of Agricultural Engineering*, 47(3), 30-35.

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- [9] De Baerdemaeker, J., Munack, A., Ramon, H., & Speckmann, H. (2001). Mechatronic systems, communication, and control in precision agriculture. IEEE Control Systems Magazine, 21(5), 48-70.
- [10] Gautam, P. V., Kushwaha, H. L., Kumar, A., & Kushwaha, D. K. (2019). Mechatronics application in precision sowing: A review. Int. J. Curr. Microbiol. App. Sci, 8(4), 1793-1807.

Motivation :

1. Bringing back the traditional method of using animals for agricultural needs.
2. Utilization of abundant animal resources in an efficient way.
3. Enhancement of the animal work output with the help of an electromechanical machine.
4. Bringing a cost effective and efficient way of farming for small to medium sized fields.
5. Decreasing the carbon footprint left by excessive use of heavy tractors by replacing them with battery operated hybrid machines.
6. Highly portable and can be used in step fields or patch fields without destroying its surroundings.

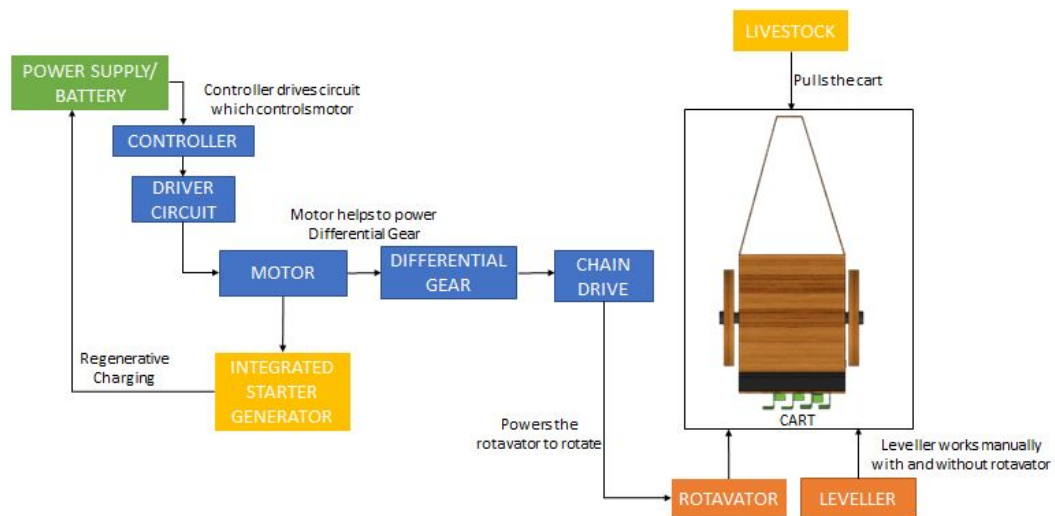
Objectives :

1. Development and fabrication of a partially automated mechanism with various agriculture tools to support varied needs in the field.
2. To promote sustainability in the agriculture sector.
3. To develop and integrate a cost-effective solution for farmers.

Organization :

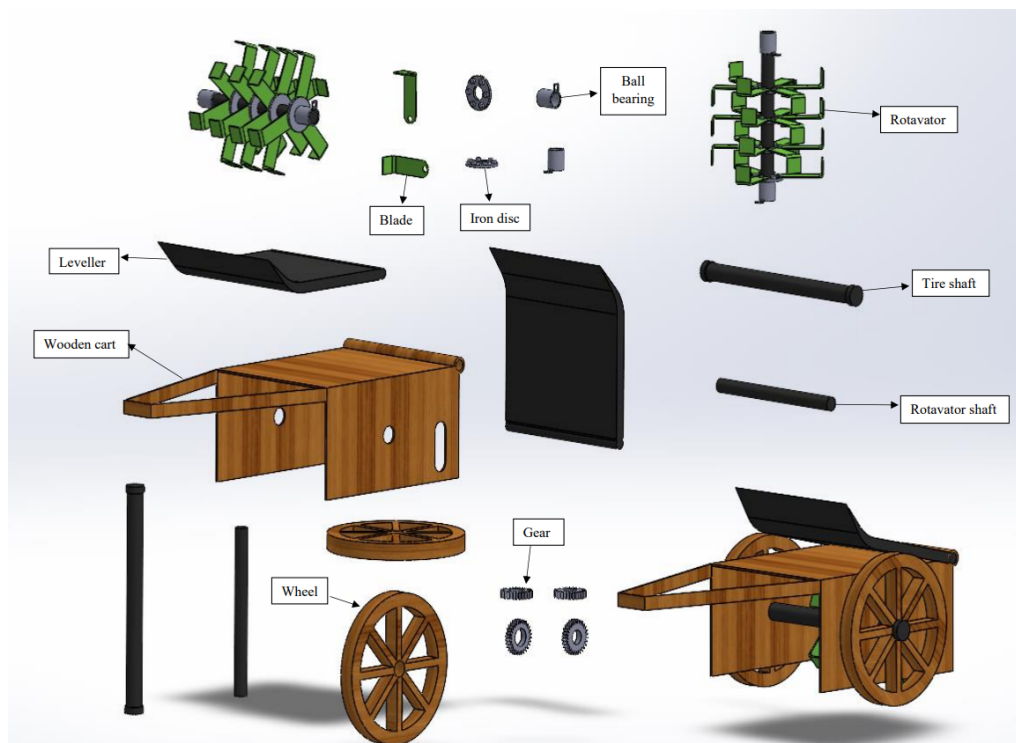
When reporting on experimental study, the whole research paper will normally include a Title page, Abstract, Introduction, Methods, Results, Discussion, and References sections. 1 Many will also include figures and tables, as well as an appendix or appendices.

Block Diagram :

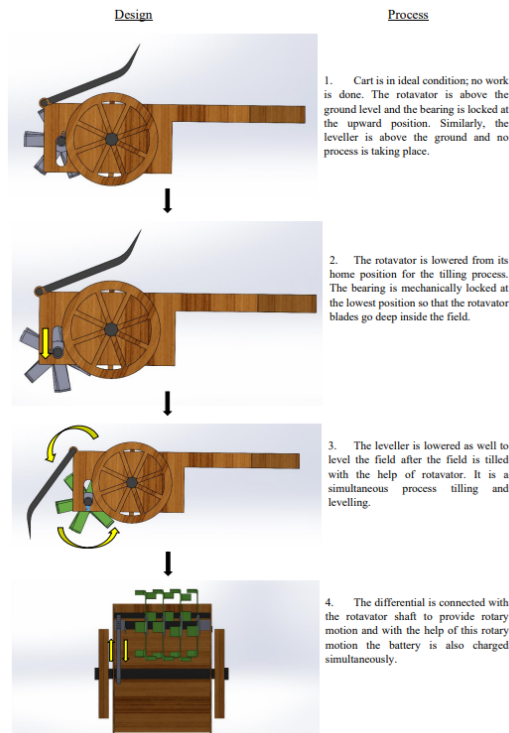


Methods :

CAD Model :



Working of Cart :



Hardware Details :

Rotavator-

Seedbed preparation is essential for sowing. A rotavator is utilized in both primary and secondary tillage. It reduces costs by reducing the amount of fuel consumed and the amount of time spent by the farmer. Soil preparation takes time, but with the appropriate tools and equipment, it can be resolved quickly. This multipurpose farming machine is a motorized machine with revolving blades that turns the soil. Rotavators are earth-turning equipment that function similarly to tillers. Rotary tillers are also ideal for weed control in locations where chemicals cannot be used due to established plants or regulatory limits. Other forms of machinery are more difficult to operate than rotavators. It can also be used to recondition crusted soil and dirt in locations prone to high rainfall or irrigation.

Land Leveler-

The tool is moved with soil towards the operator to fill tiny depressions in the seedbed. It contributes to a more efficient irrigation system, allowing all areas to be irrigated. It enhances crop establishment, crop stand, crop maturation, and field water coverage.

Furthermore, it has the potential to enhance farming land by 5 to 7%. Furthermore, it can cut operation time by 10% to 15%. It is possible to transition from planting and transplanting to direct seeding. This modification can reduce labor by 30 person-days per acre. By levelling,

you can eliminate weeds by up to 40%. Aside from that, it can enhance average yield by 10 to 20%.

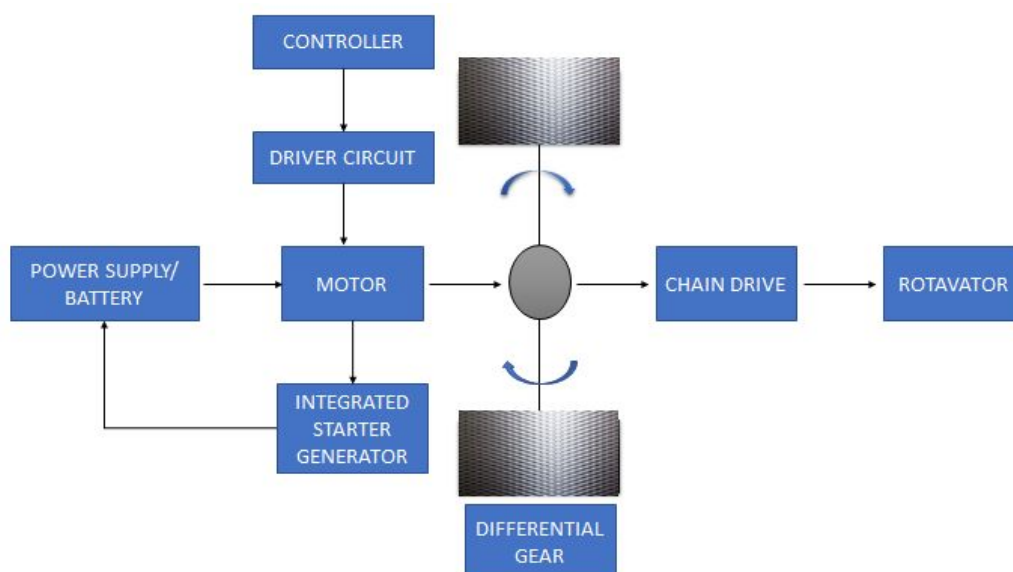
Differential Gear-

Power from the engine is sent to a pair of driving wheels, balancing the force equally between them but allowing them to follow routes of varying lengths, such as when turning a corner or navigating an uneven road. When one-wheel slips, such as when driving on mud, the torque to the other wheel is lowered, one of the axles and the ring gear are linked together by a clutch. When one wheel loses traction, the clutch resists its desire to spin, supplying more torque to the other wheel. Despite the high amount of power delivered by the transmission system, the differential limits the speed in relation to its movement to the right or left. Both drive wheels can turn at the same speed in the same direction. At different speeds, both driving wheels might turn in the same direction. Both drive wheels can rotate at the same speed in different directions. Both driving wheels can rotate at different speeds in the opposite direction.

Wooden Bullock Cart-

This cart will be less expensive than the tractor's hydraulic carts. The vertical and horizontal movement of the cart reduces the amount of man power necessary in a standard cart. The efficiency of using bullocks is determined by feeding, upkeep, yoking style, and training. It is estimated that only approximately 10% of animal energy is now utilized.

Schematic Diagram :



Result & Discussion :

Conclusion :

The majority of the power we use in our daily lives comes from nonrenewable energy sources such as petroleum and coal, which are limited in nature. Why not consider the traditional manner of Indian culture, which has long satisfied the society's energy needs? The old wisdom enabled individuals to see the big picture rather than simply the urgent remedy. By living in harmony with nature, humans may meet all of their requirements from nature in a sustainable manner. Even science agrees on the intrinsic interdependence of various life forms, which is what we have achieved between the electromechanical system and livestock.

Medium-term fertility-building leys become less economically viable when there are no cattle to generate an income. Our farming methods aim to conserve the traditional carts used in the fields by offering a small amount of assistance to the cattle via an electromechanical system. It has yielded considerable results, our development promotes expansion of convectional agricultural practices, it can help the country gain worldwide rural recognition ,has the potential for future growth and can replace expensive equipment to perform the same tasks.

The introduction of improved hand tools and improved bullock-drawn equipment marked the beginning of mechanization in Indian agriculture. Ploughing, harrowing, levelling, sowing, inter-culture, manure application, harvesting, threshing, winnowing, and other agricultural processes necessitate the use of a wide range of implements. The cost-effective solution is a one-step solution for the variety of processes carried out on the field. The implements and machinery for Indian farmers are simple in design, allowing illiterate farmers to operate them and village artisans or mechanics to maintain them.

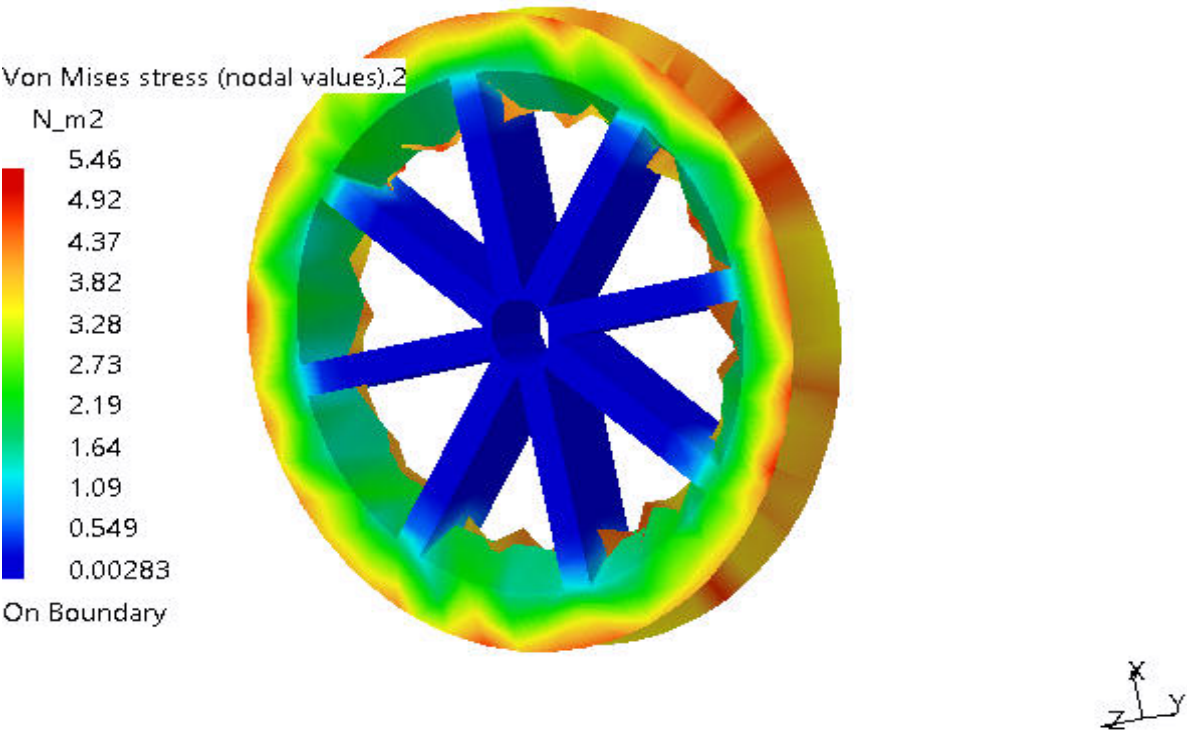
The continued use of heavy machinery on agricultural land increases soil compaction and affects soil physical qualities. Tractors and related implements are becoming increasingly expensive. The costs of operations, repairs, and maintenance are skyrocketing. Small farmers cannot afford to buy and operate tractors due to the high original cost, operational costs, and repair and maintenance costs. Traditional systems can increase soil quality by employing sustainable practices. Sustainable agriculture strives to improve a landscape's composition while delivering adequate crops.

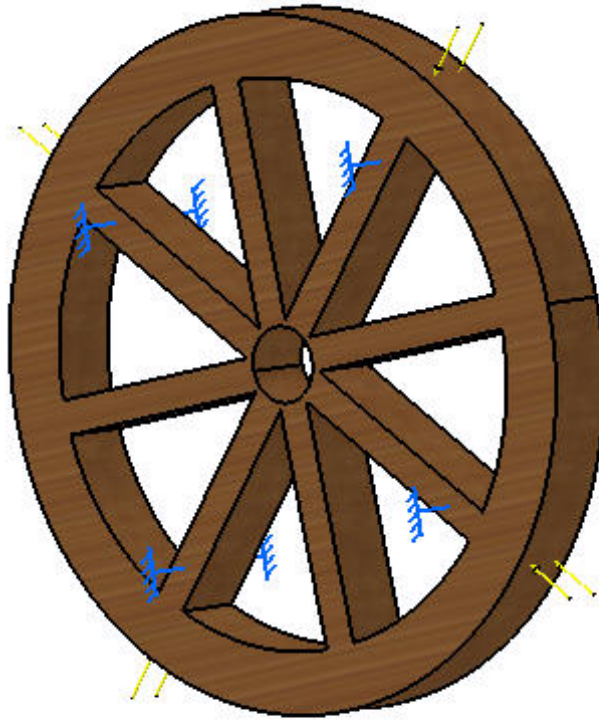
Analysis

Material	India Teak
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Young's modulus	1.6e+010N_m2
Poisson's ratio	0
Density	450kg_m3
Coefficient of thermal expansion	0_Kdeg
Yield strength	0N_m2

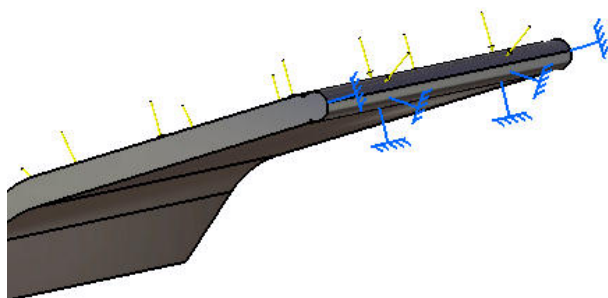
Wheel





1. Leveller

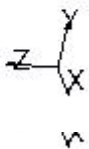
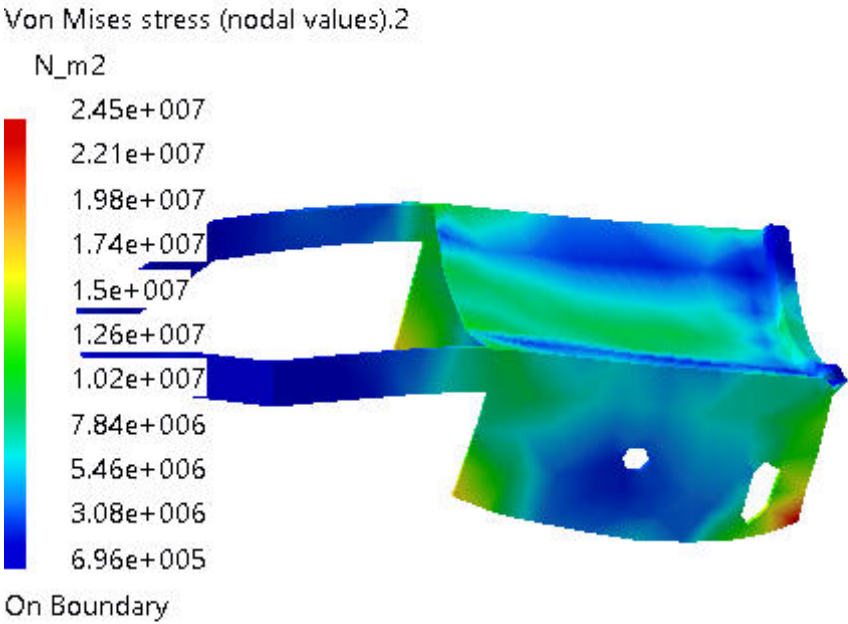
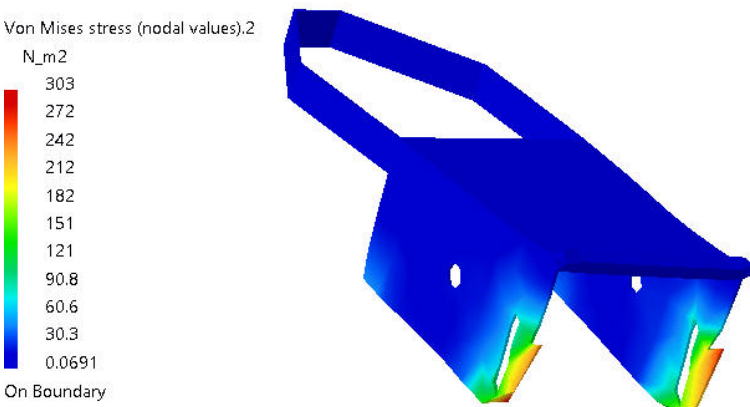
Material	Iron
Young's modulus	1.2e+011N_m2
Poisson's ratio	0.291
Density	7870kg_m3
Coefficient of thermal expansion	1.21e-005_Kdeg
Yield strength	3.1e+008N_m2



3. Cart

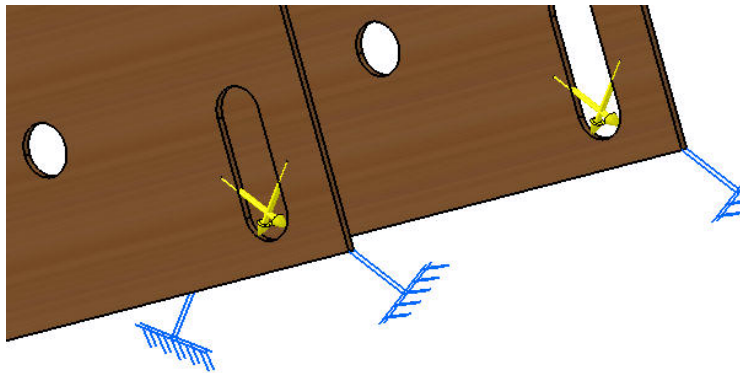
Material	India Teak
Young's modulus	1.6e+010N_m2
Poisson's ratio	0
Density	450kg_m3
Coefficient of thermal expansion	0_Kdeg

Yield strength	0N_m2
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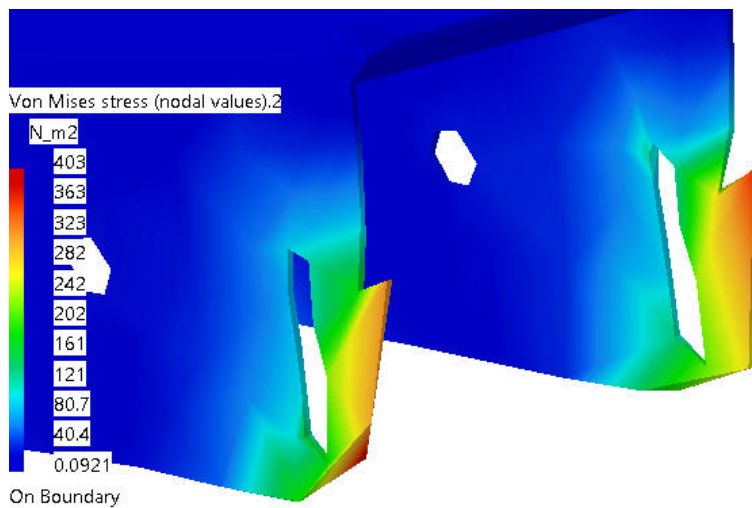


4. Cart

Material	India Teak
Young's modulus	1.6e+010N_m2
Poisson's ratio	0
Density	450kg_m3
Coefficient of thermal expansion	0_Kdeg
Yield strength	0N_m2



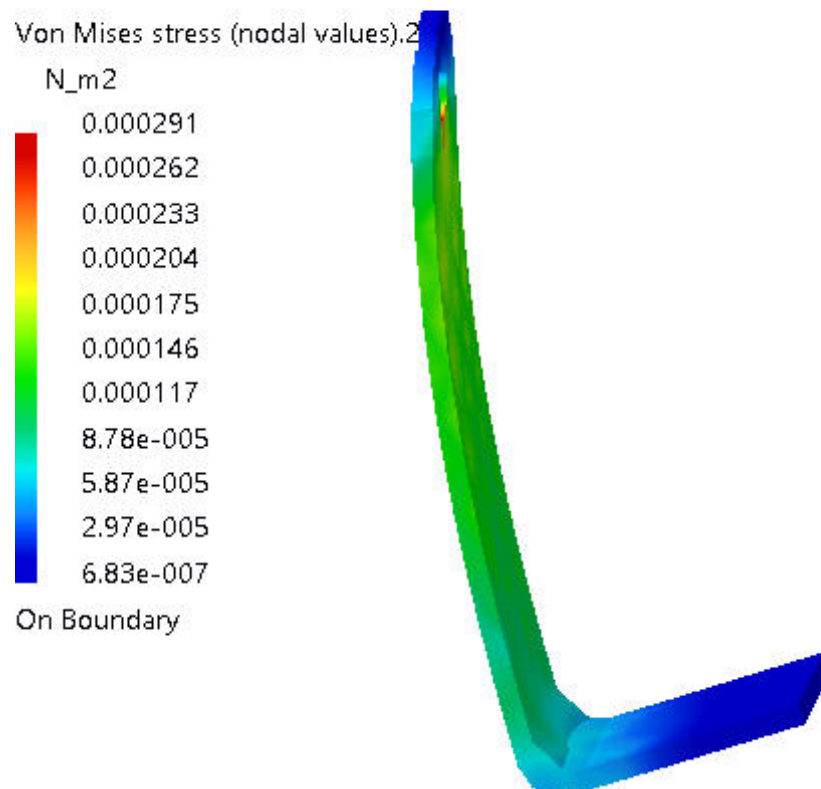
Y

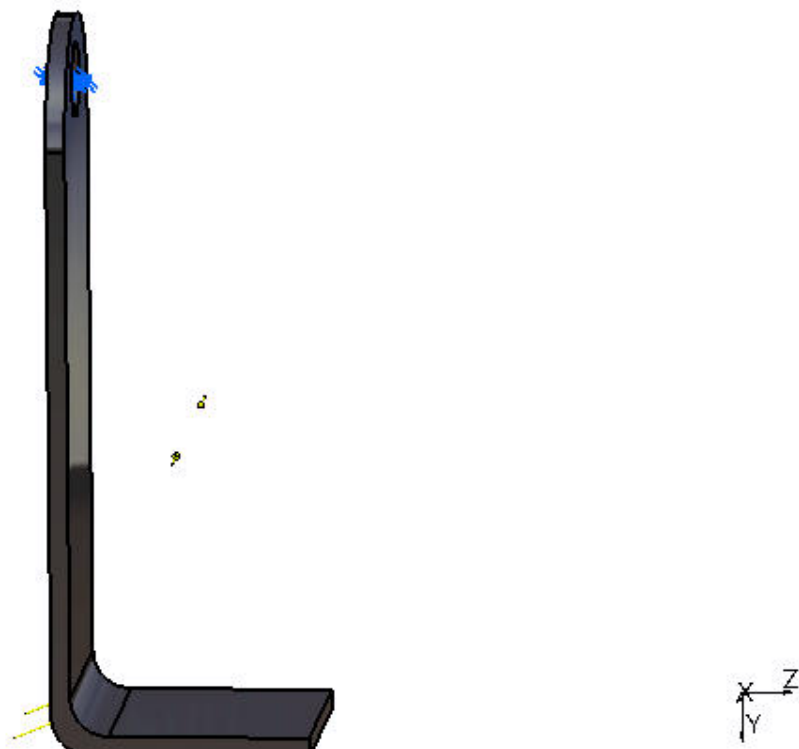


Y

5. Blade

Material	Iron
Young's modulus	1.2e+011N_m2
Poisson's ratio	0.291
Density	7870kg_m3
Coefficient of thermal expansion	1.21e-005_Kdeg
Yield strength	3.1e+008N_m2

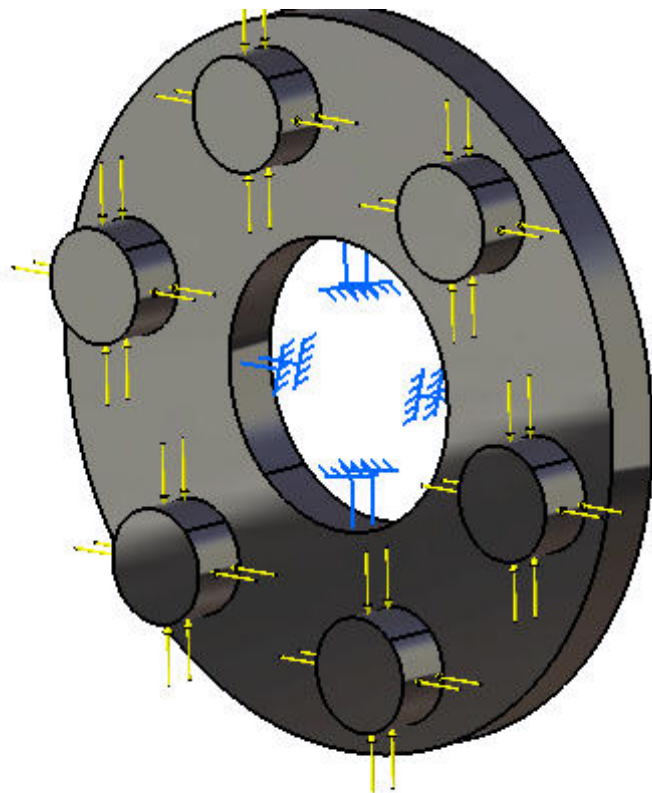




6. Rotavator plate

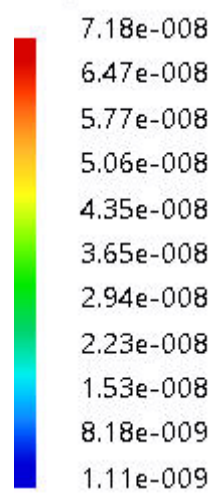
Material	Iron
Young's modulus	$1.2 \times 10^{11} \text{ N/m}^2$
Poisson's ratio	0.291

Density	7870kg_m3
Coefficient of thermal expansion	1.21e-005_Kdeg
Yield strength	3.1e+008N_m2



Von Mises stress (nodal values).2

N_m2



On Boundary

